DETAILED COMBINED RECONNAISSANCE REPORT

Networking • Passive Discovery • Gobuster • Subfinder/DNSx • DVWA

Introduction

Purpose of the Report: This report presents five mini-projects covering networking, Linux, passive information gathering, directory and subdomain enumeration, and reconnaissance of a local vulnerable web application.

Scope and Responsible Use: The exercises used `example.com` for passive reconnaissance, a local Python server for Gobuster, and a local DVWA instance at `127.0.0.1:42001`. All activities were limited to these learning targets.

Method: Each project involved identifying the target and tools, collecting results, interpreting findings, and explaining their relevance to security assessment. Screenshots provide supporting evidence.

Q1 — Networking and Linux Fundamentals

Objective: To identify the Kali Linux machine's network interface, IP address, subnet, default gateway, and DNS resolver configuration, and to test basic connectivity to the gateway.

Tools and environment: Kali Linux running in a virtual machine. Commands used: ip addr show eth0; ip route show dev eth0; ping -c 4 10.0.2.2; and inspection of /etc/resolv.conf.

Purpose: This report documents five mini-projects covering Linux networking, passive information gathering, Gobuster directory enumeration, Subfinder/DNSx subdomain enumeration, and local DVWA reconnaissance.

Scope and Responsible Use: The exercises used `example.com` for passive discovery and subdomain enumeration, a local Python HTTP server at `127.0.0.1:8000` for Gobuster, and DVWA at `127.0.0.1:42001`. Testing was limited to these learning targets.

Method: Commands and tools were used to collect evidence, interpret results, and explain their relevance to security assessment. Screenshots document the commands and findings.

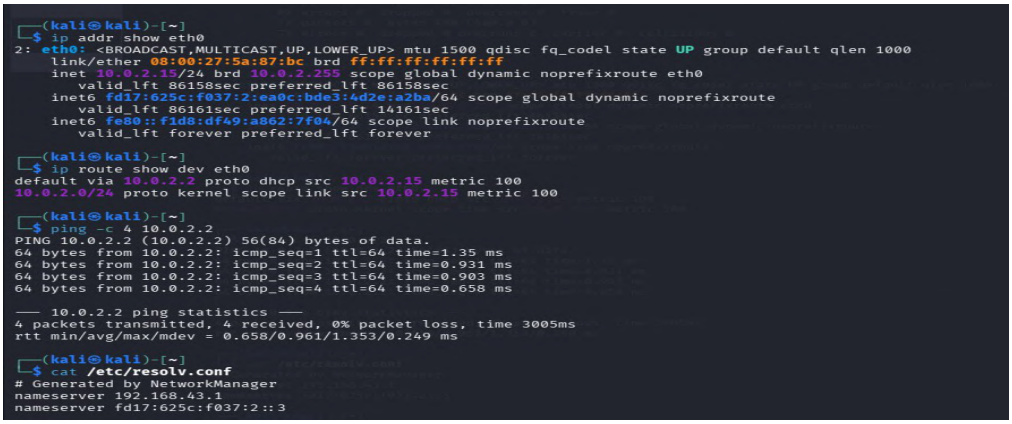


Figure 1: Network Configuration and Connectivity Verification in Kali Linux

The screenshot shows the IP address of the network interface, routing information and default gateway, ping connectivity results, and DNS resolver configuration. These commands were used to understand the network setup and verify connectivity between the Kali Linux virtual machine and its gateway.

Q2 — Information Gathering Using Browser Extensions

1. **Objective**: To identify publicly visible technologies, hosting infrastructure, IP addresses, and service information useful for authorized security assessment.

2. Target and Tools

Target: **example.com**
Tools: Wappalyzer, Netcraft, Shodan, and Kali Linux **dig**.

3. Wappalyzer — Observation and Interpretation

Finding: Wappalyzer identified Cloudflare CDN and HTTP/3 support.

Usefulness: Cloudflare indicates the use of CDN or proxy infrastructure, while HTTP/3 identifies a supported web communication protocol. These details help establish the target's technology profile.

4. Netcraft — Observation and Interpretation

Finding: Netcraft reported IPv4 address 104.20.23.154, identified Cloudflare, Inc. as the network owner and Cloudflare as the hosting provider, and displayed network, domain, nameserver, and SSL/TLS information.

Usefulness: These details help map hosting infrastructure and understand the target's domain and connection-security context. They do not independently confirm a vulnerability.

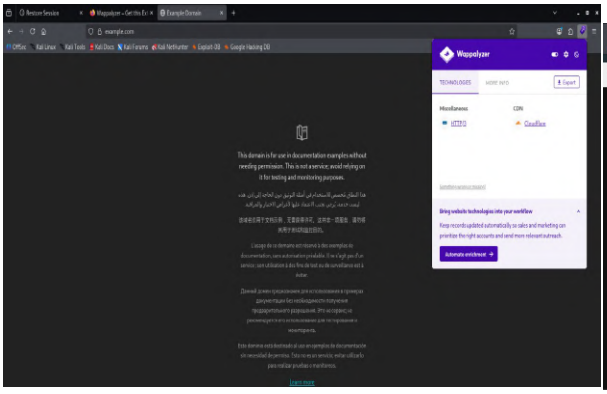
5. Shodan — Observation and Interpretation

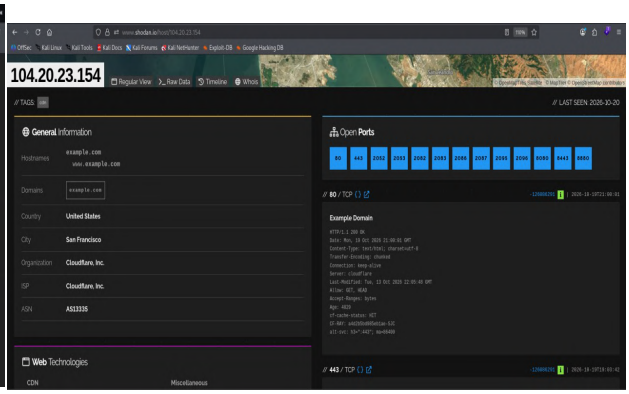
Finding: The `dig` command returned IPv4 addresses `172.66.147.243` and `104.20.23.154`. Shodan displayed associated hostnames, domain information, port listings, and Cloudflare-related server details.

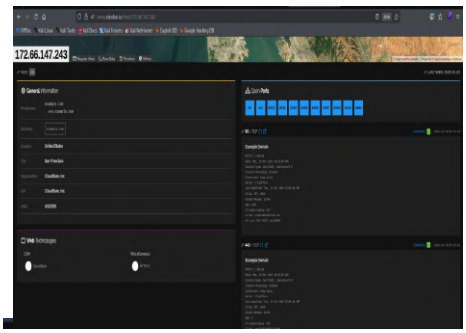
Usefulness and limitation: This information helps identify publicly observable infrastructure and services. Since the addresses are associated with Cloudflare, listed ports and services cannot automatically be attributed exclusively to the target website or treated as evidence of vulnerabilities.

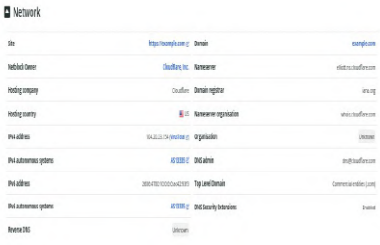
6. **Excluded Findings**: Unrelated Shodan results that merely mentioned **example.com** were excluded because no direct connection to the target was established. This avoids incorrect attribution.

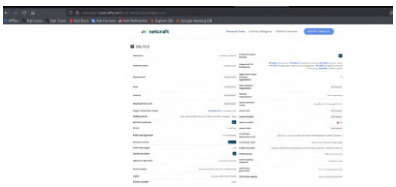
7. **Conclusion**: The exercise identified Cloudflare-related infrastructure, HTTP/3 support, hosting details, IP addresses, and publicly visible service information. Reconnaissance was completed, but no vulnerability was confirmed.











Q3 — Scenario-Based Gobuster

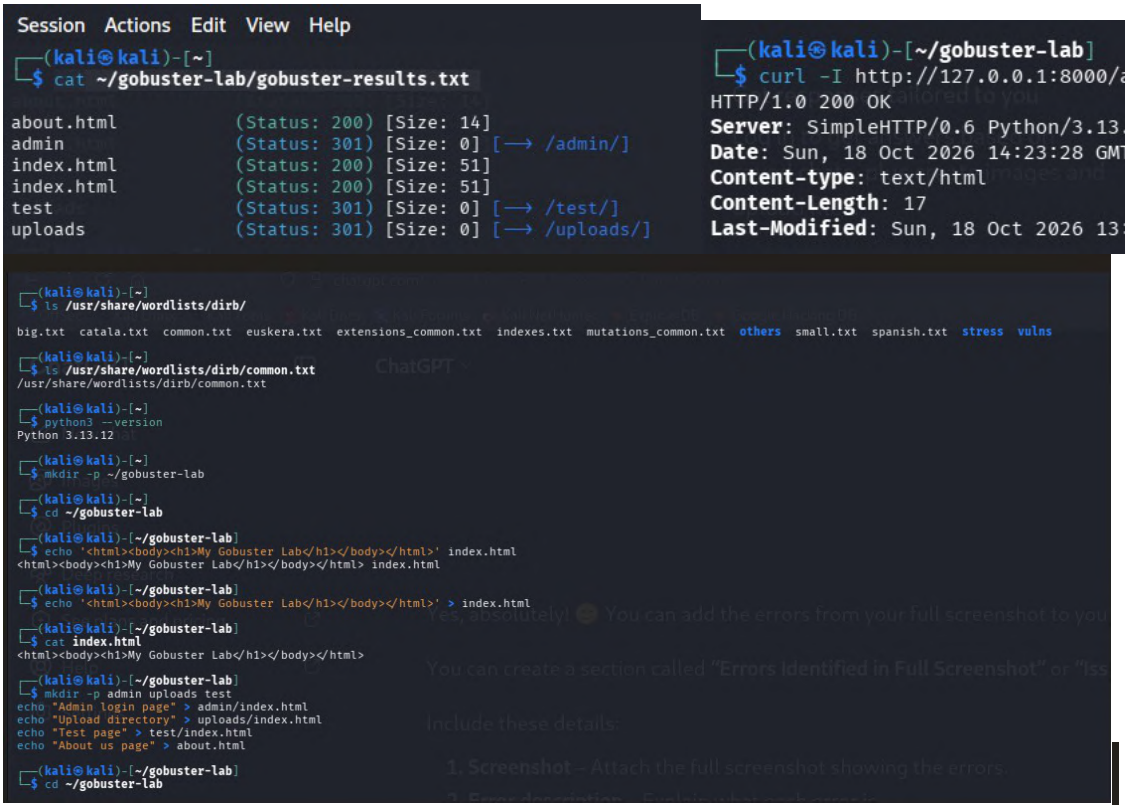
Command Used

```
curl -I http://127.0.0.1:8000
gobuster dir -u http://127.0.0.1:8000 -w /usr/share/wordlists/dirb/common.txt
```

Analysis of Findings: Gobuster identified index.html and about.html with HTTP 200 responses and /admin, /test, and /uploads with HTTP 301 redirects, as recorded in the lab results.

HTTP 200 indicates a successful response, while HTTP 301 indicates a permanent redirect. These findings help map paths available on the local test server and identify resources for further checking. A discovered path or redirect alone does not confirm a vulnerability.

Limitation: Results depend on the wordlist and server responses; undiscovered paths may still exist.



Python server running, curl -I response, Gobuster command, and complete result list.

Q4 — Scenario-Based Subdomain Enumeration

Commands Used

```
subfinder -d example.com -silent -o subdomains.txt
sort -u subdomains.txt -o subdomains.txt
cat subdomains.txt
dnsx -l subdomains.txt -silent -a -aaaa -resp -o resolved.txt
cat resolved.txt
```

Analysis of Findings: Subfinder discovered five candidate subdomains: dev.example.com, m.example.com, products.example.com, support.example.com, and www.example.com. **Discovery** identifies candidate names but does not prove they resolve.

DNSx resolved `www.example.com`, returning two IPv4 (A) addresses—`104.20.23.154` and `172.66.147.243`—and two IPv6 (AAAA) addresses—`2606:4700:10::6814:179a` and `2606:4700:10::ac42:93f3`. **Resolution** confirms that DNS returned address records for this hostname during the test.

Why It Matters: Resolved subdomains help map the target's DNS-visible infrastructure and identify hosts for further authorized assessment. The other four names were not returned by DNSx during this run; this does not prove they do not exist. Discovery and resolution alone do not confirm vulnerabilities.



Q5 — Mini Reconnaissance Project

1. **Technology Identification**
Finding/Evidence: WhatWeb identified nginx/1.30.1, PHP, HTML5, and DVWA at `127.0.0.1:42001`.
Significance: Identifies technologies relevant to further authorized checks.

2. HTTP Headers and Login
Finding/Evidence: `curl -I` returned HTTP 302 redirecting to `login.php`; the login page returned HTTP 200. Cookies included `HttpOnly` and `SameSite=Strict`.
Significance: Reveals login flow and session-cookie protections for further review.

3. Gobuster Enumeration
Finding/Evidence: Discovered `/config/`, `/database/`, `/docs/`, `/external/`, `/favicon.ico`, `/php.ini`,

`/phpinfo.php`, `/robots.txt`, and `/index.php`, with HTTP 200, 301, or 302 responses.
Significance: Maps candidate paths for further inspection; discovery alone does not prove access or vulnerability.

4. robots.txt
Finding/Evidence: Contained `User-agent: *` and `Disallow: /`.
Significance: Reveals crawler instructions, not access restrictions.

5. PHP Configuration
Finding/Evidence: `/php.ini` displayed `magic_quotes_gpc = Off`, `allow_url_fopen = on`, and `allow_url_include = on`.
Significance: These values may warrant checking, but the file does not confirm active runtime settings.

Conclusion: Reconnaissance identified technologies, login behaviour, paths, and configuration clues. DVWA was set to Impossible. No vulnerability was confirmed or exploited.

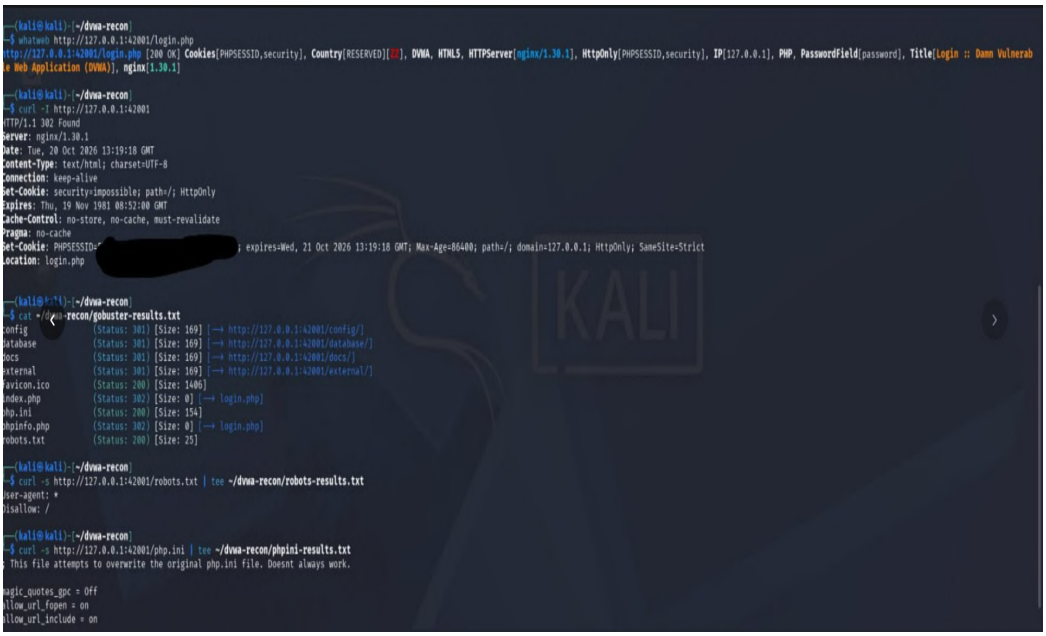


Fig:Consolidated Evidence of Web Application Reconnaissance Using WhatWeb, curl, Gobuster, and Configuration File Inspection.

Cross-Project Analysis: The five projects covered Linux networking, passive information gathering, web directory enumeration, subdomain discovery, and local DVWA reconnaissance.

The findings demonstrate how reconnaissance tools collect network, DNS, technology, and web application information. Each result must be interpreted carefully, as discovered paths may redirect, candidate subdomains may not resolve, and configuration files may not reflect active settings.Screenshots and commands provide evidence and support reproducibility. These exercises were introductory reconnaissance activities; no vulnerability was confirmed or exploited.

Final Conclusion: These projects provided practical experience with basic networking and common reconnaissance tools. They highlighted the importance of collecting evidence, interpreting findings, and understanding the limitations of tool output during a security assessment.